



Factors affecting breeding practices and surplus calf management: Producer perspectives from Atlantic Canada

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ABSTRACT

Dairy-beef and veal production play an important role in the North American beef supply, relying almost entirely on surplus calves that are marketed at a young age from dairy farms. However, suboptimal health conditions and improper management, due to their lack of a substantial role in dairy production and limited economic returns, have drawn public attention to the issue of surplus calves. To facilitate optimal care for these calves, crossbreeding and on-farm dairy-beef production have been promoted as strategies to increase the economic value of surplus calves. The objectives of this study were to describe current surplus calf marketing practices, the rationale behind these practices, and any differences between the breeding strategies employed by dairy farmers who raised and finished dairy-beef animals on their farms (referred to as “dairy-beef producers”) and those who did not (“non-dairy-beef producers”). An online survey was completed by 70 dairy producers in Atlantic Canada; 66 of these dairy producers answered sufficient questions to be classified as dairy-beef producers or non-dairy-beef producers. There was a moderate number of dairy-beef producers (24.2%) in Atlantic Canada, although most dairy-beef and non-dairy-beef producers marketed some of their surplus calves before weaning (97.0%). On-farm resource availability (i.e., housing, labor, and feeding capacities) and economic factors (i.e., production costs, profitability, quick earnings, and dairy-beef demand) were the primary considerations among producers that influenced their marketing practices. Additionally, the use of sexed frozen dairy semen (43.8%–72.0%) and unsexed frozen beef semen (78.0%–93.08%) on dairy cattle was common among producers, with Aberdeen Angus

being the most commonly used breed for crossbreeding (89.5%). The results highlight that dairy producers consider on-farm resource availability and economic factors when managing and marketing surplus calves. Crossbreeding and the use of sexed semen seem to be widely adopted on farms in Atlantic Canada, and on-farm dairy-beef production could be a viable secondary business for some dairy producers in the near future.

Key words: surplus calves, breeding, producer perspective, calf marketing

INTRODUCTION

Surplus calves, either bull or non-replacement heifer calves from dairy farms, are commonly marketed into the beef supply chain (“dairy-beef” marketed at 12–14 mo of age). Dairy-beef cattle are an important part of the beef supply in North America. In Canada, dairy-beef accounts for ~8% of the 3.5 million cattle slaughtered annually (Reed et al., 2022; Government of Canada, 2025). A 2021 survey in Ontario and Atlantic Canada showed that most dairy producers marketed surplus calves before 14 d of age, whereas some raised these calves longer before marketing (Wilson et al., 2023). The common practice of marketing surplus calves at an early age is often linked to the financial pressure of rearing costs, which limit producers’ ability to provide adequate care and keep these calves for a longer period (Wilson et al., 2023).

Marketing surplus calves at an early age negatively affects their health, particularly when producers have limited capacity to provide adequate care (Wilson et al., 2020a). In Canada, surplus calves often experience improper management on dairy farms beginning shortly after birth, such as ineffective colostrum management, inappropriate navel care, failure to administer vaccinations, and insufficient access to milk and feed (Renaud et al., 2017; Ferraro et al., 2025). These practices contribute to suboptimal health and failure of passive

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

immunity transfer (Renaud et al., 2020), increasing the risk of morbidity and mortality among surplus calves (Abdallah et al., 2022). Moreover, the limited number of calf- and beef-rearing facilities in some areas may require many surplus calves from dairy farm to be transported for long durations. For example, the National Farmed Animal Health and Welfare Council in Canada reported that surplus calves in Atlantic Canada are often transported for long periods to Québec and Ontario, where calf- and beef-rearing facilities are highly concentrated (NFAHW Council, 2019). These calves are often exposed to various stressors during transportation that further compromise their health and welfare, which does not align with public values (Creutzinger et al., 2021).

Increasing the economic returns from surplus calves has been suggested as a key factor to enhance producers' willingness to provide adequate care and encourage them to keep their calves for a longer period (Creutzinger et al., 2021). In Canada, the market price of surplus calves has tremendously increased over the past 5 years (Ferraro et al., 2025). This increase in value has also coincided with recent changes to the Canadian transport regulations requiring that calves ≤ 8 d old only be directly transported for up to 12 h without reloading and prohibit their movement through assembly centers or livestock markets. Meanwhile, calves ≥ 9 d old are allowed to be transit for up to 12 h and travel through assembly centers or livestock markets; however, unloading and reloading are permitted in order to provide feed, water, and rest if the total transport time exceeds 12 h (Government of Canada, 2020). These requirements may create a barrier for dairy producers in Atlantic Canada (Wilson et al., 2023), where transporting young surplus calves often takes >12 h. Therefore, these dairy producers may need to raise surplus calves longer before being transported, which limits their financial benefits from these calves compared with other regions due to higher rearing costs, even with potential gains from rising market prices. Moreover, market price fluctuations for surplus calves (Renaud et al., 2017) may make it challenging to consistently manage and sustain financial benefits from these calves under the current conditions in Atlantic Canada.

Dairy producers in Atlantic Canada may be more likely to find long-term solutions that enhance the economic value of surplus calves and secure financial benefits, such as crossbreeding with beef semen and rearing surplus calves for dairy-beef production (Creutzinger et al., 2021). These solutions could also help producers address the health and welfare issues associated with improper health management and marketing practices. To our knowledge, no research has described why dairy producers choose to market their surplus calves at different ages

or examined the breeding practices used on their farms. Therefore, the objectives of this study were to describe current surplus calf marketing practices, the rationale behind these practices, and any differences in breeding strategies employed by dairy farmers who raise and finish dairy-beef cattle on their farms (hereafter referred to as "dairy-beef producers") and those who choose not to ("non-dairy-beef producers").

MATERIALS AND METHODS

This cross-sectional study was an online survey approved by the Research Ethics Board at the University of Prince Edward Island (Charlottetown, Prince Edward Island, Canada; REB no. 6011910). The study was conducted between October and December 2023 using an online platform (SurveyMonkey; SurveyMonkey Inc., Palo Alto, CA). The study followed the guidelines of Strengthening the Reporting of Observational Studies in Epidemiology—Veterinary Extension (STROBE-Vet; O'Connor et al., 2016) and Consensus-Based Checklist for Reporting of Survey Studies (CROSS; Sharma et al., 2021).

Survey Development

A preliminary survey was developed by our research team using SurveyMonkey after conducting a literature review on surplus calves and dairy-beef production in Canada. As part of the development process, the survey was validated and pretested by experts specializing in dairy and beef production in Atlantic Canada. Their feedback regarding the clarity, wording, flow of the questions, and essential information regarding surplus calves was incorporated into the final version before its distribution to dairy producers. The final survey comprised 63 items within 6 sections (Supplemental File S1, see Notes), and a summary of each is provided in the sections that follow. This survey was part of a larger study assessing relevant stakeholders, marketing practices, health management, and antimicrobial use in surplus calves in the dairy-beef supply chain. Aspects of the survey used to reach the objectives of this study are described below.

Introductory Statement, Consent Form, and Contact Information (3 Items). This section was provided at the start of the survey, stating the survey background information, and confirming anonymity and confidentiality of responses and consent to participate.

Farm Characteristics (11 Items; Section A) and Producer Background Information (5 Items; Section B). To describe the study population, all participants were asked to provide information about their farm characteristics (such as location, predominant breed, primary housing for lactating cows, herd size, average milk production

per cow per day, and breeding practices, such as the use of sexed dairy or unsexed beef semen for heifers or lactating cattle), as well as demographic information (such as age, gender, education level, and number of years in dairy farming). These details were collected through a combination of multiple-choice and open-ended questions.

Stakeholders and the Dairy-Beef Production Chain (25 Items; Section C). This section included questions about stakeholders in the dairy-beef supply chain, as well as surplus calf management practices. Participants were asked how they manage their surplus calves, including practices such as newborn calf euthanasia, marketing as preweaning or weaned calves, and raising or marketing calves for veal or as finished dairy-beef steers or heifers. Finished dairy-beef steers or heifers were defined as cattle with dairy genetics (purebred or crossbred with beef) that were raised for beef production rather than milk production and marketed at 12 to 14 mo of age. Additionally, participants were asked to report the most common marketing route, the number of calves sold, and the average weight and age of calves over the 12 mo before the study period for each marketing practice that they reported. This section included a combination of question formats, including fill-in-the-blank, open-ended, dichotomous (yes or no), and multiple-choice questions, as well as ranking of selected options.

Dairy Producers' Motivations and Challenges Toward Dairy-Beef Production in Atlantic Canada (4 Items; Section D). To understand the rationale behind producers' decisions regarding their marketing practices, each participant was asked, following their responses about surplus calf marketing practices in section C, to rate their level of agreement (5-point Likert scale) with various statements related to their chosen marketing practices. Additionally, an optional open-ended question was also included to give participants the opportunity to explain the reasoning behind their practices.

Antibiotic Use in Surplus Calves on Dairy Farms (14 Items; Section E). This section was included as part of a separate study focused on understanding the dairy-beef production system and antimicrobial use among surplus calves. Participants who reported marketing their surplus calves as preweaning or weaned calves in section 4 were also asked about their current antimicrobial use practices and observed antibiotic-resistant infections in their surplus calves over the 12 mo before the study period. This section included a combination of question formats, including fill-in-the-blank, dichotomous (yes or no), multiple-choice, and ranking of selected options.

Survey Closing Statement with Consent Request for Further Contact and Additional Information (1 Item). Finally, all participants were informed about how their responses would be used and how to contact the research

team if they had any concerns regarding the research project. They were also asked to provide consent for further contact and to receive additional information.

Survey Administration

The research team contacted the provincial milk marketing boards from New Brunswick, Nova Scotia, Prince Edward Island, and Newfoundland and Labrador to distribute a web-based survey link or QR code and to send biweekly reminder messages to 519 dairy producers in the region. The research team also directly contacted participants by email if their answers were unclear or contradictory. Local veterinarians were also contacted via the Atlantic Bovine Practitioners Association to help encourage dairy producers in their area to participate in the study. As an offered incentive, all participants were entered in a drawing to win an Apple iPad. A sample size calculation was not performed due to the exploratory design of the study; however, all of dairy producers in the regions were intensively recruited through those various strategies to maximize the sample size.

Statistical Analysis

Data Management. Data were exported from SurveyMonkey to Microsoft Excel for Mac version 16.105.3 (26020123) and imported into Stata 18 (StataCorp) for data management, data visualization, and descriptive analysis. Duplicate records and multiple submissions with partial responses from the same producers were removed from the online survey dataset, and 70 records were retained in the final dataset. To achieve the objectives of this study, data from sections A through D of the survey were used. This included farm characteristics and producer background information, breeding practices, surplus calf management practices, and the participants' level of agreement with the provided rationales for their chosen marketing practices. Surplus calf marketing practices for each participant were categorized, without considering missing data, into one or more of the following: (1) selling preweaning calves, (2) selling weaned calves, and (3) selling finished dairy-beef steers or heifers. Because producers likely used more than one strategy to market their calves, they were able to select any of these choices. To better understand the marketing and breeding practices of dairy producers who chose to finished dairy-beef animals on their farm, producers were categorized as either (1) a dairy-beef producer, who had raised and sold at least one finished dairy-beef steer or heifer within the past 12 mo, or (2) a non-dairy-beef producer, who did not finish any dairy-beef animals within the past 12 mo.

Descriptive Analysis. Unanswered questions were excluded from the analysis; therefore, the number of

responses varied across analyses. To provide a comprehensive overview, the producer characteristics (i.e., dairy farm characteristics, producer demographics, and geographic regions), breeding practices, surplus calf marketing practices, and the rationale behind producers' decisions to market surplus calves among dairy-beef and non-dairy-beef producers were reported descriptively. Different analyses were performed depending on data types. The distributions of all categorical variables were described separately for dairy-beef and non-dairy-beef producers. Additionally, the proportions of respondents' level of agreement (5-point Likert scales) with their rationale behind producers' decisions for their chosen marketing practices were separately calculated for dairy-beef and non-dairy-beef producers. Chi-squared tests and Fisher's exact test were used to compare all categorical variables of producer characteristics, the number of beef semen breeds, and the most frequently used beef semen breeds on dairy farms between participants categorized as dairy-beef and non-dairy-beef producers. These tests were also used to compare the proportions of study participants using different housing systems with those reported in census data from Atlantic Canada collected in 2023 (Government of Canada, 2023) to demonstrate the representativeness of our study participants. On the other hand, to describe continuous data, measures of central tendency (mean and median) and measures of variation (percentiles, minimum, and maximum) for all continuous variables were calculated and described separately for dairy-beef and non-dairy-beef producers. Additionally, a 2-sample independent *t*-test was used to compare the average milk production per cow per day over the last 12 mo, as well as the average proportion of cattle in the herd bred with beef semen, between dairy-beef and non-dairy-beef producers. The Mann-Whitney test was used to compare the average number of lactating cows on farms.

RESULTS

Participant Demographics

A total of 70 dairy producers participated in the survey, resulting in an overall participation rate of 13.5% ($n = 70/519$). Participation rates across the 4 provinces were similar, with 14.4% ($n = 24/160$) from New Brunswick, 14.1% ($n = 20/142$) from Prince Edward Island, 12.3% ($n = 24/195$) from Nova Scotia, and 9.1% ($n = 2/22$) from Newfoundland and Labrador. Of 70 dairy producers, 69 answered the first question related to surplus calf marketing practices about whether or not they sold preweaning calves; however, due to the structure of the survey, some producers did not answer all of the questions about other marketing practices (selling weaned calves, $n = 66$;

selling finished dairy-beef steers or heifers, $n = 66$). Of the producers that completed all 3 of these questions, 24.2% ($n = 16/66$) were classified as dairy-beef producers, and the rest were non-dairy-beef producers (75.8%; $n = 50/66$). The characteristics of study participants categorized as dairy-beef and non-dairy-beef producers are presented in Tables 1 and 2. Additionally, to demonstrate the representativeness of the study population, the proportions of study participants and Atlantic Canadian dairy producers from the 2023 census using different housing systems are presented in Table 3.

Surplus Calf Marketing Practices

Overall, most dairy producers (97.0%; $n = 64/66$) reported selling some of their surplus calves in the preweaning period. The next most frequently reported practice was retaining some surplus calves and raising them as finished beef animals (24.2%; $n = 16/66$), followed by selling some surplus calves after weaning (13.6%; $n = 9/66$), and lastly, not selling any surplus calves (1.5%; $n = 1/66$). Additionally, 5 participants (7.6%; $n = 5/66$; one dairy-beef and 4 non-dairy-beef producers) reported euthanizing some newborn surplus calves. The surplus calf marketing practices among dairy-beef and non-dairy-beef producers are presented in Table 4.

Reasons Behind Surplus Calf Marketing Practices

The proportions of respondents' level of agreement with reasons for selling some of their surplus calves as preweaning or weaned calves among dairy-beef and non-dairy-beef producers are presented in Figures 1 and 2, respectively. Briefly, both dairy-beef and non-dairy-beef producers were likely to agree or strongly agree that limited housing capacity was a rationale for selling surplus calves either before or after weaning, and likely to disagree or strongly disagree that poor survival of surplus calves on dairy farms was the rationale behind their practices compared with other reasons. The proportion of dairy-beef producers level of agreement with reasons for producing dairy-beef cattle are presented in Figure 3. Dairy-beef producers generally agreed or strongly agreed that having knowledge about raising dairy-beef animals was a rationale for raising and marketing calves as finished dairy-beef steers or heifers, whereas they disagreed or strongly disagreed that government support was a rationale behind their practices compared with other reasons.

Breeding Practices

Breeding practices among dairy-beef and non-dairy-beef producers are presented in Tables 1 and 2 and Figure

Table 1. Characteristics of participants and breeding practices (categorical variables) reported by 16 dairy-beef and 50 non-dairy-beef producers participating in an online survey in Atlantic Canada in 2023

Section	Item	Percentage by producer type			P-value
		Dairy-beef producers (n = 16)	Non-dairy-beef producers (n = 50)	Total (n = 66)	
Geographic region	Province				0.929
	New Brunswick	31.3	34.0	33.3	
	Nova Scotia	43.8	34.0	36.4	
	Prince Edward Island	25.0	28.0	27.3	
	Newfoundland and Labrador	0.0	4.0	3.0	
Producers' demographics	Gender				0.486
	Male	87.5	72.0	75.8	
	Female	12.5	26.0	22.7	
	Prefer not to say	0.0	2.0	1.5	0.105
	Age				
	<30 years old	0.0	4.0	3.0	
	30–39 years old	12.5	28.0	24.2	
	40–49 years old	31.2	20.0	22.7	
	50–59 years old	56.3	26.0	33.3	
	>60 years old	0.0	18.0	13.6	
	Prefer not to say	0.0	4.0	3.0	0.140
	Education level				
	Primary, middle, or high school	6.3	28.0	22.7	
	Postsecondary education	87.5	66.0	71.2	
Dairy farm characteristics	Postgraduate education	6.35	6.0	6.1	1.000
	Owner working off dairy farms				
	Yes	6.3	6.0	6.1	
	No	93.7	92.0	92.4	
	Prefer not to say	0.0	2.0	1.5	0.712
	Primary housing of lactating cows				
	Tiestall	25.0	12.0	15.2	
	Freestall	75.0	88.0	84.9	
	Organic farm				1.000
	Yes	0.0	2.0	1.5	
	No	100.0	98.0	98.5	
	Using Lactanet				0.971
	Yes	62.5	62.0	62.1	
	No	37.5	38.0	37.9	
Breeding practices	Predominant breed in milking herd				0.242
	Holstein	93.8	100.0	98.5	
	Jersey	6.2	0.0	1.5	
	Number of beef semen breeds used				0.103
	0 breed	6.3	0.0	1.5	
	1 breed	43.7	62.0	57.6	
	2 breeds	43.7	20.0	25.8	
	3 breeds	6.3	16.0	13.6	
	Prefer not to say	0.0	2.0	1.5	0.023
	Most frequent breeds of beef semen used ¹				
	Aberdeen Angus	73.3	95.9	89.5	
	Simmental/Aberdeen Angus cross (SimAngus)	13.3	4.1	6.3	
	Limousin/Aberdeen Angus cross (Lim-Flex)	6.7	0.0	1.6	
	British Blue	6.7	0.0	1.6	

¹Respondents could enter and rank the top 3 beef breeds being used on their herds according to the frequency in the last 12 mo. The data presented are the most frequent breeds of beef semen used among 3 beef breeds. A total number of respondents was 15 dairy-beef producers and 49 non-dairy-beef producers because one dairy-beef producer reported they did not want to share the information and one non-dairy-beef producer reported they did not use any beef semen on their farms.

4. Most dairy-beef and non-dairy-beef producers reported using at least one beef breed on their farms, with ~40% of cows in their herd being bred with beef semen. Aberdeen Angus (purebred and crossbred), Limousin, British Blue, Belgian Blue, Hereford, Charolais, and Wagyu were commonly used, with Aberdeen Angus being the most popular among both dairy-beef and non-dairy-beef

producers. Additionally, unsexed frozen beef semen was more commonly used for breeding both lactating cows and heifers than other methods, with ~40% of cows in herd bred with beef semen. Likewise, sexed frozen dairy semen more commonly used among dairy-beef and non-dairy-beef producers for breeding both lactating cows and heifers than other methods.

Table 2. Characteristics of participants and breeding practices (continuous variables) reported by 66 dairy producers (16 dairy-beef and 50 non-dairy-beef producers) participating in an online survey in Atlantic Canada in 2023

Section	Item	Producer type	Minimum	Mean	Percentile			Maximum	P-value
					25th	50th	75th		
Dairy farm characteristics	Herd size (lactating cows)	Dairy-beef producers	22.0	101.1	52.0	63.5	132.5	315.0	0.120
		Non-dairy-beef producers	25.0	129.2	72.0	107.0	152.0	520.0	
		Total	22.0	122.3	58.0	100.0	150.0	520.0	
	Average milk production per cow (kg/d)	Dairy-beef producers	12.0	31.9	27.5	35.0	37.5	42.0	0.042
		Non-dairy-beef producers	21.0	35.1	32.0	35.5	38.0	44.0	
		Total	12.0	34.3	32.0	35.0	38.0	44.0	
Producer demographics	Number of dairy farming years	Dairy-beef producers	12.0	25.9	21.5	25.0	32.0	38.0	0.686
		Non-dairy-beef producers ¹	1.0	25.2	13.0	25.0	40.0	46.0	
		Total	1.0	25.1	15.0	25.0	35.0	46.0	
		Dairy-beef producers	0.0	45.8	22.5	54.0	62.5	90.0	
Breeding practices	Proportion of cows in herd bred with beef semen (%)	Non-dairy-beef producers	5.0	43.5	30.0	50.0	60.0	80.0	0.686
		Total	0.0	44.1	25.0	50.0	60.0	90.0	

¹The total number of respondents was 65 (16 dairy-beef and 49 non-dairy-beef producers) because one non-dairy-beef producer reported they did not want to share the information.

DISCUSSION

To our knowledge, this is the first study to comprehensively describe surplus calf management strategies, including marketing calves at different ages and euthanasia. Additionally, this study provides insight into the rationale behind producers' decisions to market calves at varying ages and the breeding strategies employed by dairy producers in Atlantic Canada. The survey demonstrated that most dairy farms market their surplus calves before weaning and that a moderate number of dairy producers in Atlantic Canada raise these calves for dairy-beef production. These practices are similar to results found from a national survey in 2015 reporting that most Canadian dairy producers market all of their surplus calves before 2 wk of age (Renaud et al., 2017).

Additionally, a study conducted in Atlantic Canada and Ontario from 2020 to 2021 showed that dairy producers marketed more than 50% of their surplus calves before weaning, while fewer producers raised these calves until or beyond weaning before marketing them (Wilson et al., 2023). In contrast, a survey from Australia in 2022 reported that most surplus calves on dairy farms were marketed for slaughter between 5 and 30 d of age; however, fewer of these calves were either marketed to be reared for dairy-beef production or kept for on-farm dairy-beef production (Dairy Australia, 2023). High premium prices have been reported as a major motivating factor to raise surplus male calves (Wilson et al. 2023). Additional factors, such as advice from buyers and veterinarians, have also played a crucial role in the decision-making process. Moreover, the lack of changes aligns with a high

Table 3. Proportion of study participants (n = 66) and Atlantic Canadian dairy producers from census data in 2023 (n = 307; Government of Canada, 2023) using different housing systems, demonstrating representativeness of the study population

Province	Tiestall		Freestall		P-value
	Participant % (n)	Census data % (n)	Participant % (n)	Census data % (n)	
New Brunswick	13.6 (3)	37.3 (41)	86.4 (19)	62.7 (69)	0.046
Nova Scotia	12.5 (3)	38.9 (42)	87.5 (21)	61.1 (66)	0.016
Prince Edward Island	5.6 (4)	42.5 (34)	94.4 (14)	57.5 (46)	0.180
Newfoundland and Labrador	0.0 (0)	0.0 (0)	100.0 (2)	100.0 (9)	N/A ¹
Overall	15.2 (10)	38.4 (117)	84.8 (56)	61.9 (190)	<0.001

¹N/A = not applicable; statistical test could not be performed.

Table 4. Distribution of surplus calf marketing practices reported by 16 dairy-beef producers and 50 non-dairy-beef producers participating in an online survey in Atlantic Canada in 2023¹

Type of producers	Surplus calf marketing practices	% (n)
Dairy-beef producers	As preweaning calves and finished dairy-beef steers or heifers	75.0 (12)
	As preweaning calves, weaned calves, and finished dairy-beef steers or heifers	18.8 (3)
	As finished dairy-beef steers or heifers only	6.2 (1)
Non-dairy-beef producers	As preweaning calves	86.0 (43)
	As preweaning and weaned calves	12.0 (6)
	Not marketing any surplus calves	2.0 (1)

¹All participants had the option to select more than one practice (i.e., selling as preweaning calves, weaned calves, and finished dairy-beef steers or heifers).

proportion of farmers, indicating that production costs and a lack of infrastructure are major reasons for selling preweaning calves (Figure 1).

To manage surplus calves, most dairy-beef producers in the current study used multiple marketing strategies, including selling their surplus calves either before or after weaning, or selling them as finished dairy-beef cattle. On the other hand, most non-dairy-beef producers marketed their surplus calves only before weaning, with only a few producers also marketing them after weaning. Therefore, several underlying factors may contribute to variations in marketing practices, particularly among dairy-beef producers who raise calves on the farm before marketing them as finished dairy-beef cattle. For instance, multiple

marketing strategies among dairy-beef producers may indicate that dairy-beef production on dairy farms may serve as a secondary business for dairy operations in Atlantic Canada, which is similar to a previous finding from the United States (Penn State University, 2023). In our study, we found that on-farm resource availability (i.e., housing capacity, labor capacity, and feeding capacity) and economic factors (i.e., production costs, profitability, the desire for quick earnings, and dairy-beef demand) were the primary considerations influencing surplus calf marketing practices among dairy-beef producers as well as non-dairy-beef producers.

Our findings indicated that most dairy-beef producers reported raising surplus calves for dairy-beef production

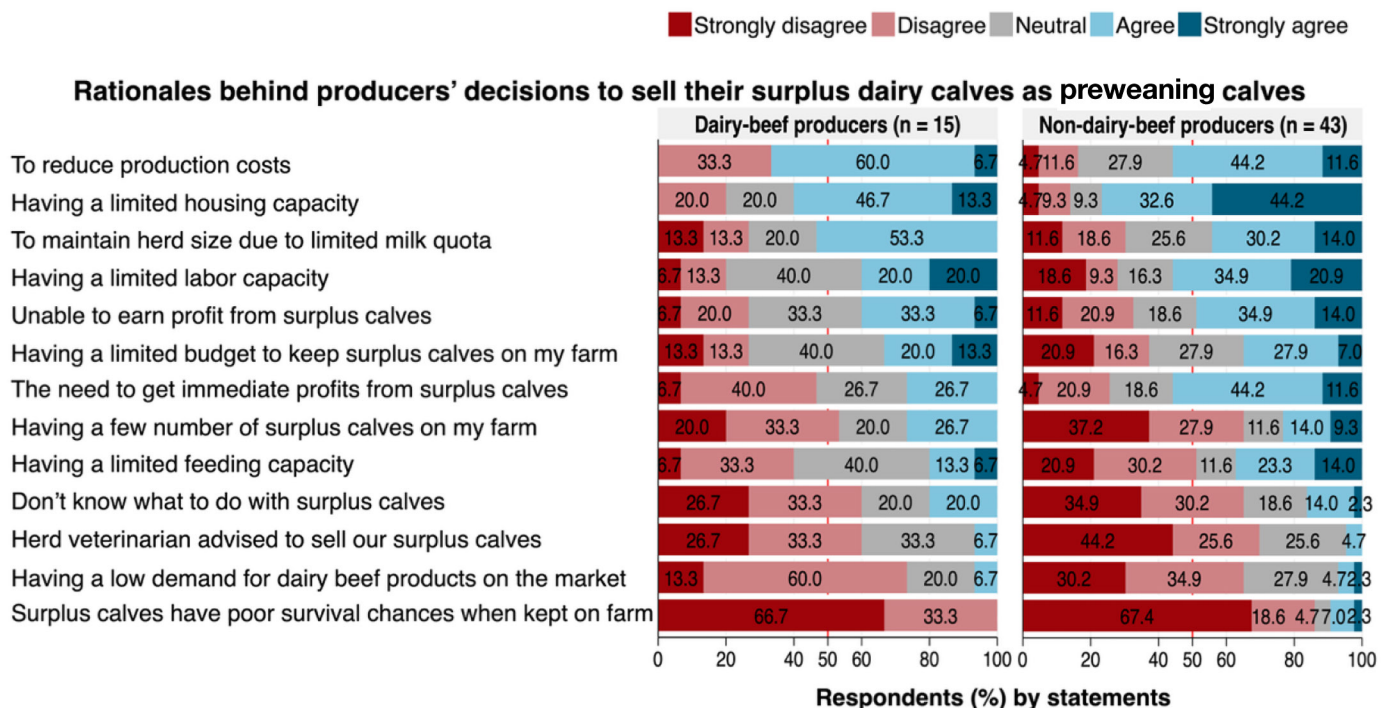


Figure 1. The proportions of respondents' level of agreement (5-point Likert scales) with 13 rationales behind producers' decisions to sell some of their surplus calves as preweaning calves among dairy farmers who also raised and finished some of their surplus calves as dairy-beef animals on their farms (dairy-beef producers; n = 15) and those who did not (non-dairy-beef producers; n = 43).

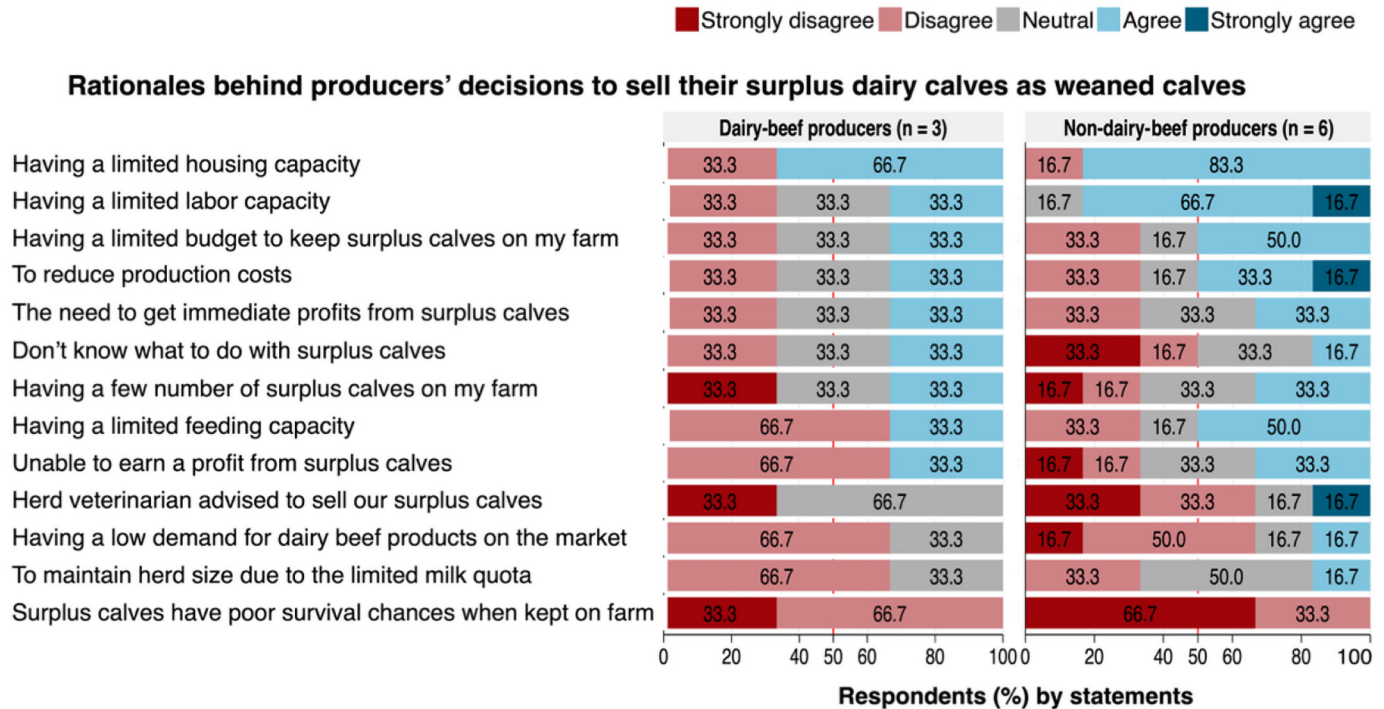


Figure 2. The proportions of respondents' level of agreement (5-point Likert scales) with 13 rationales behind producers' decisions to sell some of their surplus calves as weaned calves among dairy farmers who also raised and finished some of their surplus calves as dairy-beef animals on their farms (dairy-beef producers; n = 3) and those who did not (non-dairy-beef producers; n = 6).

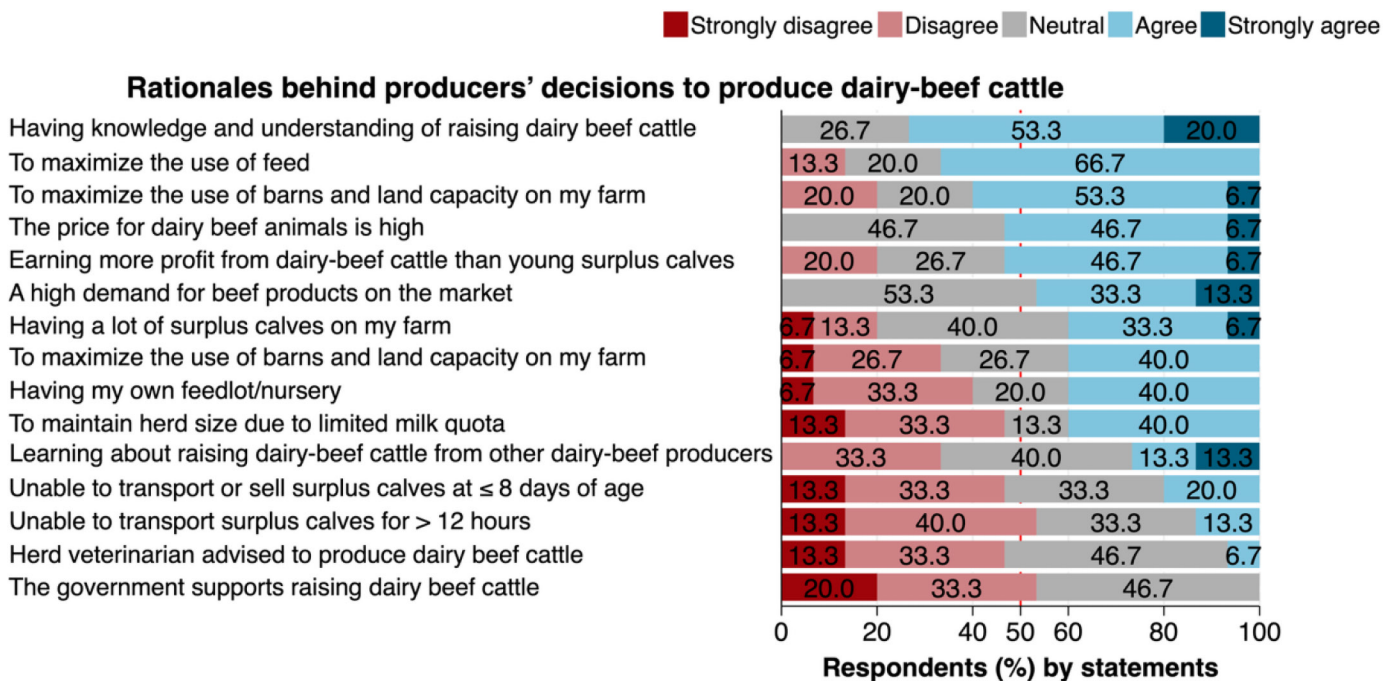


Figure 3. The proportions of respondents' level of agreement (5-point Likert scales) to rationale behind producers' decisions to sell their surplus calves as finished dairy-beef steers or heifers among 16 dairy-beef producers.

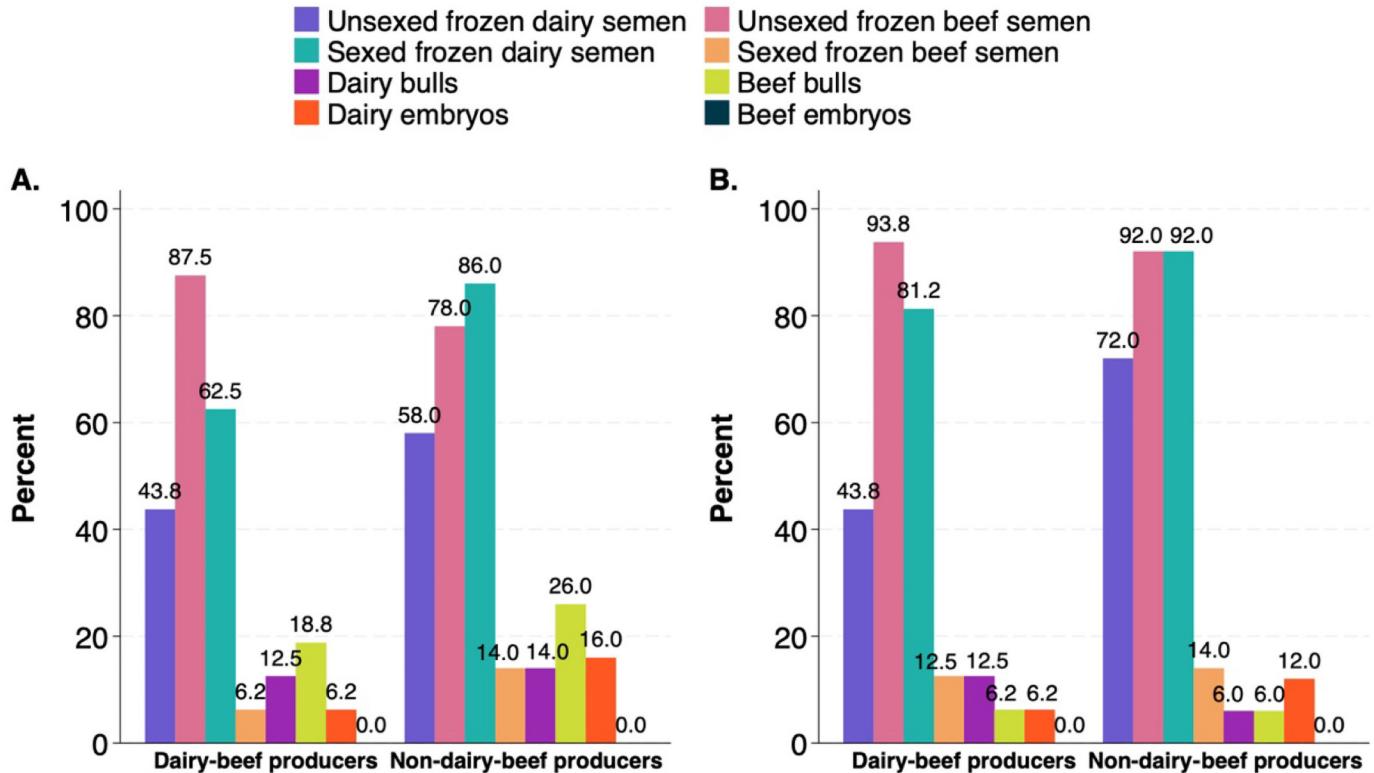


Figure 4. Breeding practices for (A) heifers and (B) lactating cows reported by dairy farmers who also raised and finished some of their surplus calves as dairy-beef animals on their farms (dairy-beef producers; $n = 16$) and those who did not (non-dairy-beef producers; $n = 50$) participating in an online survey in Atlantic Canada in 2023. Participants had the option to select more than one practice.

to maximize the use of homegrown feeds (66.7%), to use available housing (60.0%), and to take advantage of high market prices and potential profits (53.4%). In contrast, most non-dairy-beef producers reported limited housing capacity (76.8%), limited labor capacity and the need to reduce production costs (55.8%), and low profitability from raising surplus calves (48.9%) influenced their decision to market these calves early, before weaning. These results suggest that dairy-beef producers may find opportunities to diversify revenue streams by using existing farm resources to raise surplus calves, which increases the value of the calves and their financial returns. In contrast, non-dairy-beef producers (55.8%) may view the role of surplus calves in their revenue stream differently, showing a stronger preference for quick earnings (or profits) rather than waiting for more profitable returns from raising dairy-beef cattle, as dairy-beef producers do. This difference point of view in regarding financial returns from surplus calves may reflect the fact that the main purpose and stream of income are from dairy production. Raising surplus calves for dairy-beef production is optional and may typically require long-term planning and investment in additional infrastruc-

ture, labor, and feed, as also emphasized in a previous study (Wilson et al., 2020a). Additionally, the decision to raise these calves for dairy-beef production may involve a trade-off between the anticipated financial gains from finished cattle and the risk of financial losses associated with mortality before reaching finishing age. As a result, non-dairy-beef producers may be less interested in raising surplus calves and prefer immediate financial return over the long-term commitment and added effort required for dairy-beef production, whereas dairy-beef producers may be more inclined to make that investment and accept the risks of financial losses associated losing these calves before finishing age. Additionally, the desire to diversify revenue streams on dairy farms may be driven by economic pressures associated with limited milk production efficiency (Jafri et al., 2024). Although our research was not specifically designed to examine the relationship between milk production efficiency and the adoption of dairy-beef production, our findings indicated that dairy-beef producers were more likely to report lower average daily milk production per cow compared with non-dairy-beef producers. Therefore, to better understand the drivers behind diversification strategies,

further research is needed to explore how limitations in milk production efficiency influence producers' decisions to adopt dairy-beef production.

Based on our results, it appeared that although existing resources on dairy farms allow dairy-beef producers to operate dairy-beef production, these resources may not be sufficient for dairy-beef producers to keep all their surplus calves, leading to the early marketing of some surplus calves before reaching their finished weight. As presented in our study, the need to reduce production costs (66.7%), limited housing capacity (60.0%), and limited labor capacity (40.0%) were the main reasons for marketing calves before weaning among dairy-beef producers. Similarly, non-dairy-beef producers also face resource constraints on dairy farms that lead them to market surplus calves at a young age. Previous studies revealed that on-farm resource availability influences when surplus calves are marketed (Wilson et al., 2020b; Creutzinger et al., 2021). Among these resources, housing capacity has been identified as the most important consideration for when these calves are marketed (Wilson et al., 2020b; Creutzinger et al., 2021). A previous study has reported that most dairy farms can only accommodate surplus calves for a short period due to limited housing space, causing them to be marketed at an early age (Wilson et al., 2020b; Creutzinger et al., 2021). Housing space is likely limited because surplus calves share rearing facilities with replacement calves, with priority given to replacement calves because of their important role in sustaining dairy herd productivity. Although most dairy producers commonly keep nearly all their newborn heifer calves to ensure sufficient young stock to replace culled dairy cows and even exceed their farm's requirements (Mohd Nor et al., 2015), this practice may create challenges for raising surplus calves on dairy farms for extended periods. Particularly among dairy-beef producers, raising surplus calves for finished dairy-beef cattle is a long process that requires considerable space for at least 12 to 14 mo, which reduces their flexibility to accommodate additional replacement calves and results in the early marketing of surplus calves before they reach finished weight. In addition to this, labor capacity is an important factor in accommodating surplus calves, as labor on dairy farms is primarily focused on managing the milking herd (Howlett, 2018; Luby et al., 2020), which may limit their ability to care for surplus calves, particularly when rearing them for dairy-beef production. This limitation of labor may be particularly noticeable in Atlantic Canada; most dairy farms are family-owned businesses that primarily rely on family labor (Howlett, 2018; Luby et al., 2020). According to our findings, both dairy-beef and non-dairy-beef producers identified limited labor capacity in raising their surplus calves.

In addition to barn space and labor, our findings indicated that the cost of raising surplus calves was a key factor influencing both non-dairy-beef producers to sell their calves earlier and dairy-beef producers to market them before reaching their finished weight, to reduce their production cost. This finding aligns with previous research in Atlantic Canada, which revealed that surplus calves can become a burden on farms due to an increase in feeding costs (Proudfoot et al., 2022). In addition to this, raising surplus calves for dairy-beef production is a long-term investment, typically requiring 12 to 14 mo or more for the animals to reach finished weight (Creutzinger et al., 2021). Because dairy-beef production seems to be a secondary business, this extended raising period may restrict financial flexibility within the dairy operation and discourage non-dairy-beef producers from raising surplus calves on their farms. This financial constraint may also cause some dairy-beef producers to market these calves before they reach finishing weight, even though finished dairy-beef cattle tend to provide higher financial returns than younger calves. Additionally, there is a trade-off between the immediate financial return from marketing young calves during periods of high market prices and the expected financial benefits from raising them to finished cattle, which requires dairy-beef producers to bear the additional expense of feeding and raising the calve with the risk of zero financial return due to mortality, which may cause these producers to reconsider investing in raising these calves further. However, as mentioned previously, space and labor costs can influence producers' decision making about raising surplus calves.

Although on-farm resource availability and economic factors seem to be the main factors influencing surplus calf marketing practices, our study found that other factors, such as knowledge of managing dairy-beef calves and the ease of access to abattoir facilities, may also affect marketing practices. A report in 2023 highlighted that producers may need additional knowledge for the effective and profitable management of dairy-beef operations, including health, nutrition, housing, and marketing (Penn State University, 2023). Anecdotally, this idea is consistent with one participant in our study, who answered an open-ended question by stating that they were building knowledge and "testing the market" before they make a serious investment in infrastructure in the future (refer to the supplemental material [see Notes] for answers to open-ended questions). Additionally, a survey from Australia in 2022 reported that dairy producers in areas with easy access to slaughterhouses marketed most surplus calves for slaughter at a young age and that few calves were either marketed or kept on-farm for dairy-beef production (Dairy Australia, 2023). Likewise, in Alberta, where feedlots and calf rearing facilities are highly

concentrated, a 2019 report showed that most surplus calves were directly marketed at ≤ 8 d of age from dairy farms to feedlots or abattoirs, as transportation regulations allow (NFAHW Council, 2019).

One dairy-beef and 3 non-dairy-beef producers in our study reported euthanizing some newborn surplus calves on their farms. Our results were similar to findings from a previous survey conducted among dairy producers in Ontario and Atlantic Canada in 2021, which reported that only 10.0% of dairy producers euthanized at least 1% of their surplus calves (Wilson et al., 2023). Additionally, a national survey in Atlantic Canada and the other provinces in 2015 revealed that 10.0% of dairy producers euthanized their newborn surplus calves (Renaud et al., 2017). The same study also reported that newborn calf euthanasia may be routinely performed by some producers, with the proportion of surplus calves being euthanized on farms ranging from 1% to 100% (Renaud et al., 2017). Euthanasia may become a necessary choice when calves suffer from an illness or injury that negatively affects their quality of life and when treatments are ineffective. However, previous studies have also proposed other reasons for early euthanasia; for example, Renaud et al. (2017) suggested that economic factors could affect the decision to euthanize surplus calves shortly after birth because producers need to house and feed them for several days before sale, which may prevent them from earning a profit. Additionally, a lack of accessibility to purchasers (i.e., calf rearing facilities, livestock drovers, and feedlots; Renaud et al., 2017; Wilson et al., 2020a) and a lack of on-farm resource availability (i.e., housing capacity, labor capacity, and feeding capacity) could influence the decision to euthanize newborn surplus calves to avoid negative health outcomes resulting from inadequate resources (Wilson et al., 2020a). An increase in price for surplus calves, as occurred after this survey, may result in fewer producers euthanizing these animals due to financial constraints. However, the dairy industry is cautioned against the routine euthanasia of healthy surplus calves after birth, as this will likely pose challenges to the social sustainability of the dairy and beef industries in Canada (Ritter et al., 2022).

Our study found high proportions of dairy-beef and non-dairy-beef producers using sexed frozen dairy semen and unsexed frozen beef semen on their heifers and lactating cows, which aligns with the trend observed among dairy producers across Canada in 2021 (Lactanet, 2022). In Canada, dairy producers are increasingly using sexed dairy semen for cattle with high-production traits to pass on desirable genetics to their heifers, and unsexed beef semen for cattle with lower-production traits to avoid the passing of undesirable traits to their replacement heifers (Lactanet, 2022; BCRC, 2023). Additionally, the use of unsexed beef semen can become a preferred strategy for

managing the replacement herd when producers have secured sufficient replacement heifers. Our study revealed that the average proportion of dairy cows or heifers being bred with beef genetics on dairy farms was 45.8% and 41.9% for dairy-beef and non-dairy-beef producers, respectively. Raising a dairy heifer for replacement on farms costs approximately US\$3,000 (Laflamme-Michaud et al., 2025); therefore, raising only the required number of replacements can decrease rearing costs on the farm and ensure farm resources are primarily reserved for dairy operations, while also earning more financial returns from marketing crossbred surplus calves (Bitante et al., 2020). Although sexed frozen dairy semen and unsexed beef semen on dairy farms has demonstrated clear benefits in dairy farm management, the application of sexed frozen dairy semen in lactating (multiparous) cows may lead to a reduction in reproductive performance (e.g., lower conception rate, higher number of services per conception) compared with primiparous cows and heifers (DeJarnette et al., 2009; Norman et al., 2010; Karakaya et al., 2014). This reduction in reproductive performance may pose challenges for multiparous cows to achieve pregnancy within the first 120 d of lactation, which is essential for maintaining an ideal production cycle with a 12-mo calving interval. Therefore, the decision to use sexed frozen dairy semen in multiparous cows need to consider a trade-off between a higher chances of getting female calves and a decline in reproductive performance, which can potentially affect herd productivity and overall profitability.

Our study also revealed that although a variety of beef breeds (e.g., Limousin, British Blue, Belgian Blue, Hereford, Charolais, Wagyu) have been being used across different herds, Aberdeen Angus was the most popular breed used among both dairy-beef and non-dairy-beef producers. The popularity of using Aberdeen Angus semen for crossbreeding in our study aligns with a previous review, which noted that coat color is a basic phenotypic trait that can add value to crossbred calves (Basiel and Felix, 2022). Many dairy producers prefer crossbred black calves, especially those from Aberdeen Angus, because the beef industry is often willing to pay a premium for black calves at slaughter (Basiel and Felix, 2022). Additionally, Aberdeen Angus are recognized for their high carcass quality, superior meat yield, and impressive growth performance (PBQ, 2020). These characteristics contribute to the high market demand for crossbred Aberdeen Angus calves, resulting in greater financial returns compared with other breeds (Basiel and Felix, 2022). In addition to this, crossing Holsteins with smaller breeds (e.g., Aberdeen Angus) produces a desirable medium frame size preferred by slaughterhouses because it is easier to handle on the production line, while also maintaining high meat quality and yield comparable to those

of larger frame size cattle (Scasta and Hauptman, 2019; PBQ, 2020). Therefore, using sexed frozen dairy semen and unsexed frozen beef semen, especially from Aberdeen Angus, allows producers to efficiently manage the numbers of replacement and maximize economic value from surplus calves (Wilson et al., 2023), particularly when these calves are raised for dairy-beef production on dairy farms.

This study has some limitations. Although multiple strategies were implemented to increase participation (i.e., providing a clear and simple study design, sending reminders, and offering an incentive), the response rate may still be considered relatively low. However, this level of response was comparable to that of previous Canadian surveys on surplus calf marketing practices, in which response rates ranged from ~7% to 9% of Canadian dairy producers in the targeted area (Renaud et al., 2017; Wilson et al., 2023). Additionally, freestall herds were likely overrepresented in the survey when compared with census data across Atlantic Canada (Table 3). The lower number of tiestall producers in the survey might be a result of raising public concern about the welfare of animals restricted under tiestall housing systems over the past few years (Beaver et al., 2021), which may have caused some tiestall producers to be less willing to participate in research studies. Extrapolating our findings to other dairy producers in Atlantic Canada should be cautiously done due to the potential for selection bias caused by nonresponse associated with the use of a convenience sampling approach. Additionally, overlapping marketing timelines may have contributed to variation in participants' interpretation of calf age groups, particularly between preweaning and weaned stages. This limitation may have resulted in only 9 respondents reporting the marketing of surplus calves as weaned, which may constrain the interpretation of our findings.

CONCLUSIONS

Our study findings revealed that most dairy farms market their surplus calves before weaning, whereas some prefer raising them for dairy-beef production, with on-farm resource availability (i.e., housing capacity, labor capacity, and feeding capacity) and economic factors (i.e., production costs, profitability, the desire for quick earnings, and dairy-beef demand) being the main considerations in their marketing decisions. The use of sexed frozen dairy semen and unsexed frozen beef semen on dairy cattle was a common practice among dairy-beef and non-dairy-beef producers, with Aberdeen Angus being the most commonly used breed for crossbreeding. These findings suggest that most dairy-beef and non-dairy-beef producers in Atlantic Canada seek to increase their financial returns by marketing crossbred surplus calves,

while dairy-beef producers may also find opportunities to diversify revenue streams and increase the value of these calves by using farm resources to raise them for dairy-beef production.

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Nonstandard abbreviations used: N/A = not applicable.

REFERENCES

- Abdallah, A., D. Francoz, J. Berman, S. Dufour, and S. Buczinski. 2022. Association between transfer of passive immunity and health disorders in multisource commingled dairy calves raised for veal or other purposes: Systematic review and meta-analysis. *J. Dairy Sci.* 105:8371–8386. <https://doi.org/10.3168/jds.2021-21671>.
- Basiel, B. L., and T. L. Felix. 2022. Board Invited Review: Crossbreeding beef × dairy cattle for the modern beef production system. *Transl. Anim. Sci.* 6:txac025. <https://doi.org/10.1093/tas/txac025>.
- BCRC (Beef Cattle Research Council). 2023. Producing Dairy-Beef: Starting Off on the Right Hoof. Accessed Feb. 3, 2024. <https://www.beefresearch.ca/blog/producing-dairy-beef-on-the-right-hoof/>.
- Beaver, A., D. M. Weary, and M. A. G. von Keyserlingk. 2021. *Invited review:* The welfare of dairy cattle housed in tiestalls compared to less-restrictive housing types: A systematic review. *J. Dairy Sci.* 104:9383–9417. <https://doi.org/10.3168/jds.2020-19609>.
- Bittante, G., R. Negrini, M. Bergamaschi, A. Cecchinato, and H. Toledo-Alvarado. 2020. *Short communication:* Pure-breeding with sexed semen and crossbreeding with semen from double-muscled sires to improve beef production from dairy herds: Weight and value of calves. *J. Dairy Sci.* 103:5258–5262. <https://doi.org/10.3168/jds.2019-18011>.
- Creutzinger, K., J. Pempek, G. Habing, K. Proudfoot, S. Locke, D. Wilson, and D. Renaud. 2021. Perspectives on the Management of Surplus Dairy Calves in the United States and Canada. *Front. Vet. Sci.* 8:661453. <https://doi.org/10.3389/fvets.2021.661453>.

- Dairy Australia. 2023. Animal Health & Welfare on Australian Dairy Farms : Results of the Dairy Australia Animal Husbandry and Genetics Survey 2022. Accessed Apr. 1, 2025. <https://nla.gov.au/nla.obj-3198062137>.
- DeJarnette, J. M., R. L. Nebel, and C. E. Marshall. 2009. Evaluating the success of sex-sorted semen in US dairy herds from on farm records. *Theriogenology* 71:49–58. <https://doi.org/10.1016/j.theriogenology.2008.09.042>.
- Ferraro, S., M. Villettaz-Robichaud, A.-S. Perrault, J. Arsenault, Y. Chorfi, M. Costa, J. Dubuc, D. Francoz, M. Rousseau, G. Fecteau, and S. Buczinski. 2025. Dairy farm management factors associated with auction sale price of young dairy calves sold at auction markets in Québec, Canada—A cross-sectional study. *J. Dairy Sci.* 108:914–927. <https://doi.org/10.3168/jds.2024-24890>.
- Government of Canada. 2023. Dairy Barns by Type. Accessed Feb. 5, 2024. <https://agriculture.canada.ca/en/sector/animal-industry/canadian-dairy-information-centre/statistics-market-information/farm-statistics/barns-type>.
- Government of Canada. 2020. Health of Animals Regulations: Part XII: Transport of Animals—Regulatory Amendment. Interpretive Guidance for Regulated Parties. Accessed Feb. 3, 2024. <https://inspection.canada.ca/animal-health/terrestrial-animals/humane-transport/health-of-animals-regulations-part-xii/eng/1582126008181/1582126616914>.
- Government of Canada. 2025. Number of Cattle Slaughtered in Provincially Inspected Establishments in Canada. Accessed Dec. 9, 2025. <https://agriculture.canada.ca/en/market-information-system/rp/index-eng.cfm?action=gR&r=111&signature=3E72ACB74212B2666ECF9DA77A97F1BF&pdctc=&pTPl=1#wb-cont>.
- Howlett, A. 2018. Labor allocation—Where is time spent on the dairy? The Manager. Progressive Dairyman. Cornell University, Ithaca, NY.
- Jafri, S. H., K. M. M. Adnan, S. Baimbill Johnson, A. A. Talukder, M. Yu, and E. Osei. 2024. Challenges and Solutions for Small Dairy Farms in the U.S. *Agriculture (Basel)* 2369. <https://doi.org/10.3390/agriculture14122369>.
- Karakaya, E., G. Yilmazbas-Mecitoglu, A. Keskin, A. Alkan, U. Tasdemir, J. Santos, and A. Gumen. 2014. Fertility in Dairy Cows After Artificial Insemination Using Sex-Sorted Sperm or Conventional Semen. *Reprod. Domest. Anim.* 49:333–337. <https://doi.org/10.1111/rda.12280>.
- Lactanet. 2022. Insemination Trends: The Rise in Sexed and Beef Sire Semen. Accessed Feb. 3, 2024. <https://lactanet.ca/en/insemination-trends/>.
- Laflamme-Michaud, L., R. A. Molano, D. E. Santschi, S. Binggeli, D. Warner, O. Brassard, S. Jetté-Nantel, E. R. Paquet, and E. Charbonneau. 2025. Characterization of heifer rearing costs and their association with farm factors of Québec Holstein dairy farms. *Can. J. Anim. Sci.* 105:1–11. <https://doi.org/10.1139/cjas-2024-0154>.
- Luby, C. D., C. L. Waldner, and M. D. Jelinski. 2020. Ownership structure and financial status of Canadian dairy operations. *Can. Vet. J.* 61:891–893.
- Mohd Nor, N., W. Steeneveld, M. C. M. Mourits, and H. Hogeveen. 2015. The optimal number of heifer calves to be reared as dairy replacements. *J. Dairy Sci.* 98:861–871. <https://doi.org/10.3168/jds.2014-8329>.
- NFAHW (National Farmed Animal Health and Welfare Council) Council. 2019. The Marketing of Male Dairy Calves in Canada. Accessed Feb. 2, 2024. http://www.ahwcouncil.ca/pdfs/animal-welfare-system/NFAHW%20Council_Recommendation_National%20Farm%20Animal%20Welfare%20System%20for%20Canada_2019.pdf.
- Norman, H. D., J. L. Hutchison, and R. H. Miller. 2010. Use of sexed semen and its effect on conception rate, calf sex, dystocia, and stillbirth of Holsteins in the United States. *J. Dairy Sci.* 93:3880–3890. <https://doi.org/10.3168/jds.2009-2781>.
- O'Connor, A. M., J. M. Sargeant, I. R. Dohoo, H. N. Erb, M. Cevallos, M. Egger, A. K. Ersbøll, S. W. Martin, L. R. Nielsen, D. L. Pearl, D. U. Pfeiffer, J. Sanchez, M. E. Torrence, H. Vigre, C. Waldner, and M. P. Ward. 2016. Explanation and elaboration document for the STROBE-Vet statement: Strengthening the reporting of observational studies in epidemiology—Veterinary extension. *J. Vet. Intern. Med.* 30:1896–1928. <https://doi.org/10.1111/jvim.14592>.
- PBQ (Les Producteurs de bovins du Québec). 2020. The Use of Beef Sire Semen in Dairy Herds: A Practical Guide. Les Producteurs de bovins du Québec.
- Penn State University. 2023. Understanding Dairy-Beef Production. Accessed Apr. 2, 2025. <https://www.thecattlesite.com/articles/understanding-dairy-beef-production>.
- Proudfoot, K. L., J. Hendricks, A. Higgins, S. Roche, C. Ritter, D. L. Renaud, and M. A. G. Von Keyserlingk. 2022. The Entrepreneurs: Dairy farmer perspectives on finding an industry solution for the surplus calf issue—A participatory case study. *Front. Sustain. Food Syst.* 6:961068. <https://doi.org/10.3389/fsufs.2022.961068>.
- Reed, L. M., D. L. Renaud, and T. J. DeVries. 2022. Male dairy calf welfare: A Canadian perspective on challenges and potential solutions. *Can. Vet. J.* 63:187–193.
- Renaud, D. L., T. F. Duffield, S. J. LeBlanc, D. B. Haley, and D. F. Kelton. 2017. Management practices for male calves on Canadian dairy farms. *J. Dairy Sci.* 100:6862–6871. <https://doi.org/10.3168/jds.2017-12750>.
- Renaud, D. L., M. A. Steele, R. Genore, S. M. Roche, and C. B. Winder. 2020. Passive immunity and colostrum management practices on Ontario dairy farms and auction facilities: A cross-sectional study. *J. Dairy Sci.* 103:8369–8377. <https://doi.org/10.3168/jds.2020-18572>.
- Ritter, C., M. J. Hötzel, and M. G. von Keyserlingk. 2022. Public attitudes toward different management scenarios for “surplus” dairy calves. *J. Dairy Sci.* 105:5909–5925. <https://doi.org/10.3168/jds.2021-21425>.
- Scasta, J. D., and B. S. Hauptman. 2019. Beef Cow Size: Industry Trends, Economics, and Implications for Grazing Wyoming Rangelands. Accessed Sep. 26, 2025. <https://wyoextension.org/publications/html/B1343/>.
- Sharma, A., N. T. Minh Duc, T. Luu Lam Thang, N. H. Nam, S. J. Ng, K. S. Abbas, N. T. Huy, A. Marušić, C. L. Paul, J. Kwok, J. Karbwang, C. de Waure, F. J. Drummond, Y. Kizawa, E. Taal, J. Vermeulen, G. H. M. Lee, A. Gyedu, K. G. To, M. L. Verra, É. M. Jacqz-Aigrain, W. K. G. Leclercq, S. T. Salminen, C. D. Sherbourne, B. Mintzes, S. Lozano, U. S. Tran, M. Matsui, and M. Karamouzian. 2021. A Consensus-Based Checklist for Reporting of Survey Studies (CROSS). *J. Gen. Intern. Med.* 36:3179–3187. <https://doi.org/10.1007/s11606-021-06737-1>.
- Wilson, D. J., D. Canning, T. Giacomazzi, K. Keels, R. Lothrop, D. L. Renaud, N. Sillett, D. Taylor, H. Van Huigenbos, B. Wynands, D. Zuest, and D. Fraser. 2020a. Hot topic: Health and welfare challenges in the marketing of male dairy calves—Findings and consensus of an expert consultation. *J. Dairy Sci.* 103:11628–11635. <https://doi.org/10.3168/jds.2020-18438>.
- Wilson, D. J., J. A. Pempek, T.-Y. Cheng, G. Habing, K. L. Proudfoot, C. B. Winder, and D. L. Renaud. 2023. A survey of male and female dairy calf care practices and opportunities for change. *J. Dairy Sci.* 106:703–717. <https://doi.org/10.3168/jds.2022-22238>.
- Wilson, D. J., J. Stojkov, D. L. Renaud, and D. Fraser. 2020b. Short communication: Condition of male dairy calves at auction markets. *J. Dairy Sci.* 103:8530–8534. <https://doi.org/10.3168/jds.2019-17860>.

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